

Mathematical Physics I (Fall 2026): Homework #1

Due Sep. 18, 2026 (Fri, 23:00pm)

[0.5 pt each, total 5 pts, turn in as a single pdf file to eTL]

- By turning in your homework, you acknowledge that you have not received any unpermitted aid, nor have compromised your academic integrity during its preparation. (Remember the SNU College of Natural Sciences Honor Code, and the SNU AI Guideline!)
- Textbook problem numbering convention (Boas, 3rd ed.): [Section].[Problem No.]. Many problems in this assignment are from the list of suggested problems announced in the class.
- Exhibit all intermediate steps to receive full credits. Only handwritten answers are accepted except for numerical problems or verifications — for which you print out and turn in not just the end results (e.g., plots) but also the source codes in your favorite tools such as MATLAB or MATHEMATICA.

1.-3. Boas Chapter 1, Problems 13.19, 16.1, 16.9

(Note: For Problem 13.19, first prove the equality given, for which you may need to prove and utilize $\sec \theta = \frac{d(\sec \theta + \tan \theta)/d\theta}{\sec \theta + \tan \theta}$. For Problem 16.1, it is helpful to draw a diagram that considers the balance between forces and torques, a so-called *extended free-body diagram*.)

4.-7. Boas Chapter 2, Problems 10.24, 12.27, 14.24, 17.19

(Note: For Problem 14.24, compare your result with a computer solution. Resolve or explain any disagreement in the two results.)

8. Boas Chapter 3, Problem 2.13

(Note: For Problem 2.13, read the instruction in the textbook carefully; among others, you have been asked to compare your result with a computer solution as well.)

9. Two coherent waves of the same angular frequency arrive at a detector. In wave optics, the two waves can be described as $y_1(t) = A_1 \cos(\omega t)$ and $y_2(t) = A_2 \cos(\omega t + \phi)$, with amplitudes A_1 and A_2 and a phase difference ϕ . Rather than adding the waves directly using trigonometric identities, one can represent each wave by a complex quantity and take the real part ($\Re[\cdot]$) only at the very end.

(a) Show that the combined total wave can be expressed as

$$y(t) = \Re \left[\left(A_1 + A_2 e^{i\phi} \right) e^{i\omega t} \right],$$

and that the amplitude of the resulting wave becomes $\sqrt{A_1^2 + A_2^2 + 2A_1 A_2 \cos \phi}$.

(b) Show that the intensity of the total wave can then be written as $I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \phi$, where I_1 and I_2 are the intensities of the two individual waves. Find the maximum and minimum possible intensities, $I_{\max}$ and $I_{\min}$, in terms of I_1 and I_2 as the phase difference ϕ varies.

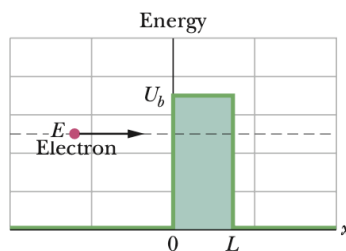
(c) Suppose that the two waves have equal amplitudes, $A_1 = A_2$. Show that your result in (b) reduces to $I = 4I_1 \cos^2 \left(\frac{\phi}{2} \right)$. Explain physically why the intensity can become four times the intensity of the individual wave. Finally, suppose that the two waves have amplitudes in the ratio $A_2 = \frac{1}{2}A_1$. Determine the ratio $\frac{I_{\max}}{I_{\min}}$.

(Note: You may want to review the freshman physics textbooks such as Halliday & Resnick.)

10. In quantum mechanics, consider the transmission coefficient T of an electron matter wave (of mass m and energy E) incident on a rectangular potential barrier of height U_b (see figure). Briefly explain why the following expression must be evaluated to determine the probability of the electron being transmitted through the barrier and detected on the other side:

$$T^{-1} = \left| \frac{(k + i\kappa)^2 - (k - i\kappa)^2 e^{-2\kappa L}}{4ik\kappa e^{-\kappa L} e^{-ikL}} \right|^2 = 1 + \left(\frac{k^2 + \kappa^2}{2k\kappa} \right)^2 \sinh^2(\kappa L), \quad (1)$$

where $k \equiv \frac{\sqrt{2mE}}{\hbar}$ and $\kappa \equiv \frac{\sqrt{2m(U_b - E)}}{\hbar}$ are real numbers. Prove the second equality using complex algebra.



[Adapted from Halliday & Resnick, Chapter 38.9]

(Note: You may want to review the freshman physics textbooks such as Halliday & Resnick (Chapters 38.8 and 38.9), or the quantum mechanics textbooks such as Griffiths & Schroeter (Chapter 2.6 and problems therein) or Landau & Lifshitz. To receive full credit, you should include a brief explanation — without a detailed derivation — of how the absolute square of a complex number like Eq.(1) arises in calculating the transmission probability.)